

and actuator control. Importantly, certain components within each zone—advanced LED lighting systems and radar modules—still have their own intelligence for performance or safety reasons.

In fact, Tier-1 suppliers often deliver these smart modules with built-in software because they possess deep, specialised knowledge (for example, how to drive a matrix LED headlamp or interpret radar signals in real time). This balance (and handshake) between centralised computing and localised intelligence is critical: it reduces wiring and complexity yet preserves the specialised autonomy of parts that need it.

Edge computing in IoT. In the broader IoT and industrial world, edge computing has become a key trend. By 2027, it will become common for even small embedded systems to include some AI or decision-making capability on-device. Drones analyse video feeds onboard to avoid obstacles in real time. Industrial sensors do local signal processing to detect anomalies without waiting for cloud analytics. This ‘intelligence at the edge’ is enabled by new chips that combine traditional microcontroller units (MCUs) with AI accelerators or efficient software frameworks for TinyML that can run on microcontroller-class hardware.

AI in the MCU. A decade ago, it was unimaginable to run a neural network on a coin-sized microcontroller running on a few milliwatts of power. Today, it is not only possible but it is also increasingly standard for certain applications. Innovations in hardware (such as ARM’s Cortex-M architectures with DSP and ML extensions, or novel low-power AI chips) and software (such as TensorFlow Lite for Microcontrollers) enable what the industry calls Embedded AI, ARM’s or Edge AI. For example, a smart HVAC system might include an MCU that uses a tiny neural network

to learn your usage patterns and optimise climate control, all locally.

The integration of AI into edge electronics means the part is no longer just following a script, it is making judgments.

This distributed intelligence has two powerful implications. First, it makes systems more resilient and responsive: a local controller can react instantly to a safety issue or optimise performance without a round-trip to the cloud or central unit. Second, it changes the design process: instead of one monolithic software stack, you have many interacting smart subsystems. Each smart part might be designed by a different specialist team or supplier, so the overall product development becomes more about system integration and coordination of these mini-brains.

Micro interactions: HMI moves to the edge

When intelligence lives in each part, **how do humans interact with those parts?** In the past, if a product had one central ‘brain,’ it often presented one unified interface: for instance, a single touchscreen on an appliance or a dashboard in a car. Now imagine a scenario where not just the product, but its components are also smart. Each one might need to signal information or be configured, or respond to users in some way. This is driving a rise in embedded HMIs at the part level, and what we might call micro-interactions.

From invisible to visible. Each smart part, by definition, has something to say—whether it is a status, a warning, or a need for input/confirmation. We see this clearly in industrial and automotive contexts: for example, a collaborative robotic arm in a factory, which used to be just an actuator, now comes equipped with status lights and makes sounds. When a human enters its workspace, it might flash a warning or issue an alert tone. These visual and audible indicators

APLUS VOICE SPEECH MELODY IC COB MODULE

- 10 SECONDS TO 176 MINUTES VOICE IC MODULE SOLUTION AVAILABLE
- MODULE FROM 10 SECONDS - 180 MINUTES
- RE-RECORDABLE VOICE FROM 10 SECONDS - 16 MINUTES
- AP89XX SERIES - 10-42 SECONDS, 3KXX SERIES - 10-42 SECONDS
- AM58Q6-7-8 10 SEC TO 60 MINUTES + IC AND MODULE AP89XXX 10 SECONDS -11 MINUTES WITH MCU INTERFACE AMTP32M-11 MINUTES AND 22 MINUTES VARIANT AVAILABLE
- AIVR MCU+LED+VOICE IC 10 SECS TO 341 SECS MANTRA
- SHLOKA IC & MODULE 10 SECOND - 3 HOURS
- AMTP32MK 11MINS AND AMTP64M - 88 MINS VOICE IC
- SAME D88 MODULE 10 SECOND TO 180 MINUTES +

CALL US NOW!
91-98333 74818

APPLICATION:

- Voice Announcement
- Fire Alarm ● Siren ● Timer ● Elevator
- Doorbell ● Token Display Voice IC in Various Languages ● Car Reverse Horn
- Automotive ● Greeting Cards ● Elevator Floor Voice Annunciator IC
- Telecom Announcement ● Telecom Music on Hold ● Telecom IVRS Voice IC
- Motor Pump IVRS IC ● Mantra Shloka Japam ● Custom Design Voice Module
- Clock Chime ● Clock Alarm ● Security Based Voice IC ● Dog Barking/Animal Sound Voice IC ● Gift And Toy Industrial Announcement Voice IC ● Islamic
- Safar Dua ● Islamic Azan ● Christian Vasanam ● LED Flasher And
- Many Other Applications Depending Upon Customers Requirement.



S M SEMICONDUCTORS

Ground Floor Bakul Baug, Vasundhara, Opp. Indian Bank, S. V. Road, Santacruz (W), Mumbai - 400 054.
M: 91-98333 74818,
E: gopalnarayan777@gmail.com

www.aplusindia.net